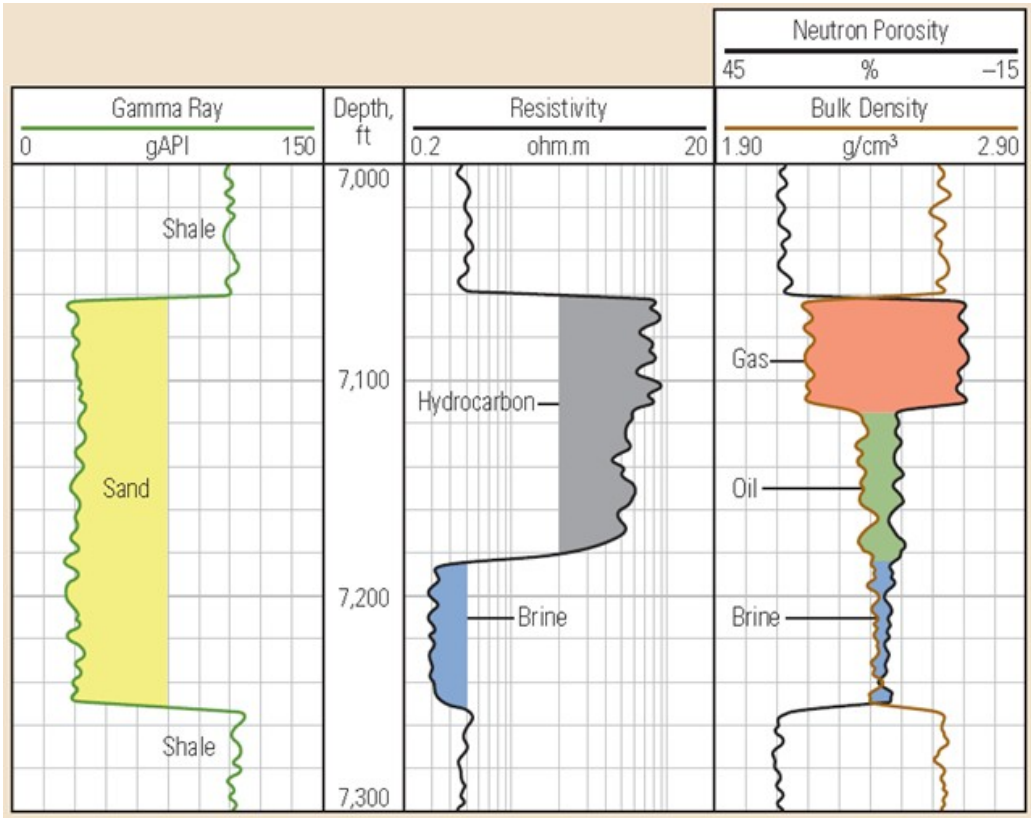
Assignment 1 - Day 3

```
from IPython.core.display import Image

Image('01-Well-logging.png')
```




```
GR = float(input("What is GR Value ? "))

if GR < 0 or GR >200:
    print("Invalid Value")

elif GR > 45:
    print("Shale")
elif GR <= 45:
    Res = float(input("What is Resistivity Value ? "))
    if Res <2:
        print("Brine Sand")
    elif Res >=2:
        Density = float(input("What is Density Value ? "))
        Neutron = float(input("What is Neutron Value ? "))
        if Density <2.2 and Neutron <15:
            print("Gas Sand")
        elif Density >=2.2 and Neutron >=15:
            print("Oil Sand")
```

```
Gas Sand
```

Assignment 2 - Day 3

```
Gradient  = 1.09
Pressure = int(input("What is the pressure value? "))

while Pressure < 5000:
    Pressure = Pressure*Gradient
    Gradient = Gradient + 0.1

    print(f"Pressure value is {Pressure} psi")
```

```
Pressure value is 3815.0000000000005 psi
Pressure value is 4539.850000000001 psi
Pressure value is 5856.406500000003 psi
```

```
i=0
while i<5:
    print(i)
    i+=1
```

```
0
1
2
3
4
```

```
for i in range(0,5):
    print(i)
```

```
0
1
2
3
4
```

```
for i in range(5):
    print(i)
```

```
0
1
2
3
4
```

```
for i in range(0,5,1):
    print(i)
```

```
0
1
```

```
2
3
4

Res_pres = 3000
Max_press = 5000

while Res_pres < Max_press:
    print(f"Pressure is {Res_pres}")
    Res_pres+=250
```

```
Pressure is 3000
Pressure is 3250
Pressure is 3500
Pressure is 3750
Pressure is 4000
Pressure is 4250
Pressure is 4500
Pressure is 4750
```

```
Res_pres = 3000
Max_press = 5000

for pres in range(Res_pres, Max_press, 250):
    print(f"Pressure is {pres}")
```

```
Pressure is 3000
Pressure is 3250
Pressure is 3500
Pressure is 3750
Pressure is 4000
Pressure is 4250
Pressure is 4500
Pressure is 4750
```

```
Res_pres = 3000
Max_press = 5000

for pres in range(Res_pres, Max_press, 250):
    print(f"Pressure is {pres}")
```

```
Pressure is 3000
Pressure is 3250
Pressure is 3500
Pressure is 3750
Pressure is 4000
Pressure is 4250
Pressure is 4500
Pressure is 4750
```

```
Res_pres = 3000
Max_press = 5000

for pres in range(Max_press, Res_pres, -250):
    print(f"Pressure is {pres}")
```

```
Pressure is 5000
Pressure is 4750
Pressure is 4500
Pressure is 4250
Pressure is 4000
Pressure is 3750
Pressure is 3500
Pressure is 3250
```

```
Res_pres = 3000
Max_press = 5000
incre = 250

for pres in range(Res_pres, Max_press, incre):
    print(f"Pressure is {pres}")
```

```
Pressure is 3000
Pressure is 3250
Pressure is 3500
Pressure is 3750
Pressure is 4000
Pressure is 4250
Pressure is 4500
Pressure is 4750
```

```
start = 0
total =0

while start <=10:
    total =total+start
    start = start + 1
print(total)
```

```
55
```

```
total = 0

for start in range(0,11):
    total =total+start
print(total)
```

```
55
```

```
n = int(input('Enter the number: '))
start = 0
```

```
total =0

while start <=n:
    total =total+start
    start = start + 1
print(total)
```

```
5050
```

```
n = int(input('Enter the number: '))
total = 0

for start in range(0,n+1):
    total =total+start
print(total)
```

```
5050
```

```
#calculate total seismic travel time if seismic wave travelling though
4 layers

layers = int(input("Enter the number of layers"))
total_travel_time = 0

for layer in range(1, layers+1):
    thicknes = int(input(f'Enetr the thickness value for layer
{layer}: '))
    velocity = int(input("Enter the velocity value for layer {layer}:
"))
    travel_time = (2*thicknes)/velocity

    total_travel_time += travel_time
print(f"Total Travel time is {total_travel_time:.2f} seconds")
```

```
Total Travel time is 2.37 seconds
```

```
#calculate total seismic travel time if seismic wave travelling though
4 layers

layers = int(input("Enter the number of layers"))
total_travel_time = 0
layer = 1

while layer <=layers:
    thicknes = int(input('Enetr the thickness value: '))
    velocity = int(input("Enter the velocity value: "))
    travel_time = (2*thicknes)/velocity

    total_travel_time += travel_time
```

```
    layer +=1
print(f"Total Travel time is {total_travel_time:.2f} seconds")
```

```
Total Travel time is 1.46 seconds
```

Netsed if inside for loop

```
Sw = int(input("What is the water saturation? "))


while Sw <=100:
    print(f"Water saturation is {Sw}")
    Sw += 10
```

```
Water saturation is 5
Water saturation is 15
Water saturation is 25
Water saturation is 35
Water saturation is 45
Water saturation is 55
Water saturation is 65
Water saturation is 75
Water saturation is 85
Water saturation is 95
```

```
Sw = int(input("What is the water saturation? "))


while Sw <=100:
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")

    Sw += 10
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

```python
Sw = int(input("What is the water saturation? "))


for i in range(Sw,101, 10):
    print(f"Water saturation is {Sw}")
```

```
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
Water saturation is 5
```

```python
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

Keywords

```python
if = "Reservoir Pressure"
```

```
  Cell In[50], line 1
    if = "Reservoir Pressure"
       ^
SyntaxError: invalid syntax
```

```python
else = 4000
```

```
  Cell In[51], line 1
    else = 4000
    ^
SyntaxError: invalid syntax
```

```
Else = 4000
```

Break Keyword

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    break
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
    break
```

```
Accpted SW: 5
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
        break
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
```

```
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
        break
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
        break
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    elif Sw == 75:
        break
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
```

```
High SW: 55
High SW: 65
```

Continue

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    continue
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
        continue
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        continue
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

```python
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
    continue
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

```python
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
        continue
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 75
Rejected Sw: 85
Rejected Sw: 95
```

```
Sw = int(input("What is the water saturation? "))

for Sw in range(Sw,101, 10):
    if Sw <=40:
        print(f"Accpted SW: {Sw}")
    elif 70>= Sw > 40:
        print(f"High SW: {Sw}")
    elif Sw ==75:
        continue
        print(f"SW is {Sw}")
    else:
        print(f"Rejected Sw: {Sw}")
```

```
Accpted SW: 5
Accpted SW: 15
Accpted SW: 25
Accpted SW: 35
High SW: 45
High SW: 55
High SW: 65
Rejected Sw: 85
Rejected Sw: 95
```

```
# porosity value from input
#If value 0.2 - 0.6 >> takes the value and stop
#if value different >> then its invalid, ask again for input

while True:
    porosity = float(input("What is the value of porosity ? "))

    if 0.6 > porosity > 0.2:
        print(f"Thanks for the input. porosity value store:
{porosity}")
        break
    else:
        print("Invalide value. Enter the value again")
```

```
Invalide value. Enter the value again
Invalide value. Enter the value again
Invalide value. Enter the value again
Thanks for the input. porosity value store: 0.5
```

Functions P1

```
#Python built in functions
#type(), print()

#Python built in Modules
#math.sqrt(), statistics.stdev()
```

```
#Define Function

#def function_name():
    #return ??
    #task/Statement to execute

#out_of_function

#Calling function

#function_name()

def greet():
    name = input("Enter your name")
    print(f'hello  {name}')

greet()
```

```
hello  Ahmed
```

```
#Define Function

#def function_name(param1, param2, param3):
    #return ??
    #task/Statement to execute

#out_of_function

#Calling function

#function_name(argument1, argument2, argument3)

def multiplication(a, b):
    return a*b

multiplication(25, 65)
```

```
1625
```

```
Sw = float(input("What is the Water Saturation value? "))

print(f"Hydrocarbon Saturation is {1-Sw}")
```

```
Hydrocarbon Saturation is 0.7
```

```
Sw = float(input("What is the Water Saturation value? "))


def HC_Sat():
    Hc = 1-Sw
    return  Hc
```

```
HC_Sat()
```

```
0.7
```

```
Sw = float(input("What is the Water Saturation value? "))


def HC_Sat():
    Hc = 1-Sw
    print(Hc)
```

```
HC_Sat()
```

```
0.7
```

```


def HC_Sat(SW):
    return f'Hydrocarbon Saturation is {1-SW}'
```

```
HC_Sat(0.3)
```

```
'Hydrocarbon Saturation is 0.7'
```

```
def porosity_check():
    while True:
        porosity = float(input("What is the value of porosity ? "))

        if 0.6 > porosity > 0.2:
            print(f"Thanks for the input. porosity value store:
{porosity}")
            break
        else:
            print("Invalide value. Enter the value again")
```

```
porosity_check()
```

```
Thanks for the input. porosity value store: 0.3
```

```
porosity_check()
```

```
Invalide value. Enter the value again
Invalide value. Enter the value again
Invalide value. Enter the value again
Thanks for the input. porosity value store: 0.3
```